1. (4 points) For an oscillator we found $\omega = \pm\omega_o$. If it started with $x(0) = A$ and $\dot{x}(0) = \sqrt{3}\omega_o A$ then find $x(t)$.

2. (6 points) A block of mass m is connected to a spring, the other end of which is fixed. There is also a viscous damping mechanism. The following observations have been made on this system:

   1. If the block is pushed horizontally with a force equal to mg, the static compression of the spring is equal to h.
   2. The viscous resistive force is equal to mg if the block moves with a certain known speed u.

   (a) (3 points) For this complete system, write the differential equation governing horizontal oscillations of the mass in terms of m, g, h, and u.
   Answer the following for the case that $u = 3\sqrt{gh}$:

   (b) (1 point) What is the angular frequency of the damped oscillation?

   (c) (2 points) If the oscillator is driven with the force $mg\cos\omega t$ where $\omega = \sqrt{2g/h}$ what is the amplitude of the steady state response?

3. (5 points) While moving, a mass m experiences a resistive force $-m\gamma v$ where v is the velocity, and γ is a constant, but no spring-like restoring force.

   (a) (2 points) Show that its displacement as a function of time is of the form

   $$x = C - \frac{v_0}{\gamma}e^{-\gamma t},$$

   where, C and v_0 are constants.

   (b) (1 point) At $t = 0$ the mass is at rest at $x = 0$. At this instant a driving force $F = F_0\cos\omega t$ is switch on. Find the values of A and δ in the steady state solution $x = A\cos(\omega t - \delta)$.

   (c) (2 points) Write down the general solution [the sum of parts (a) and (b)] and find the values of C and v_0 from the conditions that $x = 0$ and $\frac{dx}{dt} = 0$ at $t = 0$, and $\omega = \gamma$.

4. (5 points) An object of mass 0.2 kg is hung from a spring who spring constant is 80 $\mathrm{Nm^{-1}}$. The body is subjected to a resistive force given by $-bv$, where v is its velocity in $\mathrm{ms^{-1}}$ and $b = 4$ $\mathrm{Nm^{-1}s}$.

   (a) (3 points) Set up the differential equations of motion for free oscillations of the system and find the period of such oscillations.

   (b) (2 points) The object is subjected to a sinusoidal driving force given by $F(t) = F_0\sin\omega t$, where $F_0 = 2$ N and $\omega = 30\ \mathrm{rad\,s^{-1}}$. In the steady state, what is the amplitude of the forced oscillation?

1. Consider a beaded string with bead mass m, bead separation a, number of beads N whose both ends are fixed.
   (a) Show that $N+1-r$ normal mode frequency ($1 \leq r \leq N$) is the same as $N+1+r$ normal mode frequency. [2]
   (b) How are the corresponding amplitudes are connected? [2]

2. A continuous string has 100 Hz difference between its first and the third normal mode frequencies. If the mass per unit length of the string is doubled and the length of the string is reduced to its three fourth value, what will be the difference of frequencies of its two lowest normal modes? [4]

3. (a) Write Maxwell's equations in matter. [2]
   (b) Show that for an EM wave traveling along z direction, E field must be in the XY plane. [2]
   (c) For the same wave, show that $\vec{B} = \frac{1}{c}(\hat{k} \times \vec{E})$. [4]

4. Show that the wave equation [4]

$$\frac{\partial^2 f}{\partial t^2} = v^2 \frac{\partial^2 f}{\partial x^2}$$

admits solution in the form $f(x \pm vt)$.

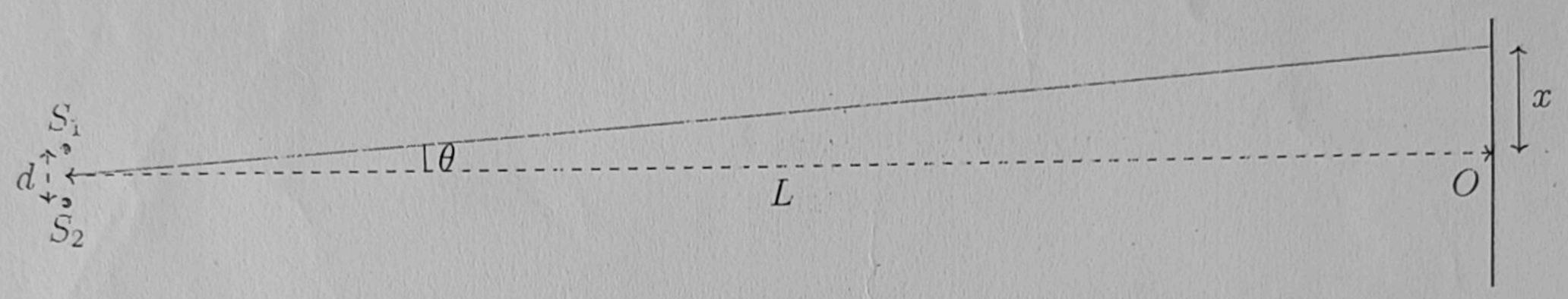
1. (a) Suppose $Ae^{iax} + Be^{ibx} = Ce^{icx}$ for some nonzero constants A, B, C, a, b, c, and for all x. Prove that $a = b = c$ and $A + B = C$. [2+2]

   (b) For a normal incidence, calculate the exact reflection and transmission coefficients without assuming $\mu_1 = \mu_2 \approx \mu_\circ$. You should start from Fresnel's equations (for *p-type*) and take appropriate limits to get normal incidence. Confirm that $I_R + I_T = I_\circ$. [2+2]

2. The refractive index of the glass is 3/2. Consider an air-glass interface and a linearly polarized *p-type* wave is incident on the interface. What is the angle at which the reflected and the transmitted beams have the same intensities? [4]

3. Consider two coherent (no phase difference between them) point sources, S_1 and S_2, kept d distance apart and a screen placed at L distance from the point sources, as shown in figure. The sources emit monochromatic light of wavenumber $k = \frac{2\pi}{\lambda}$. The intensity of light at position x is given by [8]

$$I(\theta) = I_\circ \cos^2\left(\frac{kd\sin\theta}{2}\right)$$



Assuming, $x \ll L$ and the entire setup is kept in air (refractive index 1), answer the following.

(a) Express the intensity as a function of x. [2]

(b) What is the distance between successive minima? [2]

(c) If we place a thin slab of glass of thickness $\delta\ell$ and refractive index η in front of S_2, calculate the resulting intensity pattern. [4]